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pH-regulated inner-sphere coordination for selective Fe(III) removal over rare-earth elements from acidic leachates using a D2EHPA-based resin

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ABSTRACT

Efficient removal of iron (Fe) from acidic leachates is essential for improving water quality and minimizing interference during downstream separation processes for rare-earth elements (REEs). However, the molecular basis governing the preferential removal of Fe(III) over REEs in di-(2-ethylhexyl) phosphoric acid (D2EHPA)-based systems remains unclear. In this work, D2EHPA was immobilized within a solid resin matrix to combine its metal-coordination capability with the operational advantages of solid-phase adsorption and fixed-bed processing. Using neodymium (Nd) as a representative REE, we combine batch adsorption experiments, interfacial characterization, and density functional theory (DFT) calculations to elucidate the underlying mechanism. The selectivity arises from pH-dependent Fe(III) speciation and inner-sphere coordination between metal ions and D2EHPA. At pH ≈ 2.5, the coexistence of Fe^{3+} and emerging $FeOH^{2+}$ species promotes adsorption, where $FeOH^{2+}$ forms electroneutral 2:1 Fe–D2EHPA inner-sphere complexes (e.g., $[Fe(OH)(D2EHPA)_2]$) that govern selective Fe removal. Zeta potential and isoelectric point shifts reveal interfacial charge redistribution, while XPS and FTIR confirm P-O-Fe bonding. DFT calculations show that Fe-D2EHPA complexes possess stronger interaction energies and greater charge transfer than Nd analogues, further promoted by hydrogen-bonded water. Kinetic and isotherm analyses indicate chemisorption-controlled, monolayer adsorption on uniform sites. Fixed-bed column tests under flow validate the mechanism, exhibiting strong Fe retention, efficient acid desorption, and stable multi-cycle regeneration. These findings identify pH-regulated inner-sphere Fe–O(P) coordination as the molecular origin of Fe-over-REE selectivity and establish a scalable strategy for Fe removal coupled with REE enrichment from complex leachates, providing a mechanistic foundation for selective Fe pre-removal in critical mineral recovery.

1. Introduction

The global transition to clean energy is fundamentally reshaping raw-material demand [1]. Low-carbon technologies—from electric vehicles to wind turbines—require far greater quantities of minerals than fossil-based systems, placing unprecedented pressure on the supply of critical minerals [2]. Among them, rare earth elements (REEs) have emerged as indispensable enablers of advanced energy and defense technologies [3]. Defined as the 15 lanthanides together with Sc and Y, REEs are essential for permanent magnets in EV motors and wind turbines, catalysts in petroleum refining, phosphors in electronics, and precision components in defense systems [4]. Recent projections indicate that REE demand will more than double by 2040 under net-zero scenarios, underscoring the urgency of developing secure and diversified supply chains [5].

At the same time, iron (Fe)—a fundamental component for steelmaking and energy infrastructure—is typically present at extremely high levels in acidic leachates generated from many sources like coal-based materials, which is one of the most promising unconventional sources of REEs [6,7]. In this context, Fe does not act as a resource but rather as a dominant impurity: it competes with REEs for extractants, consumes separation reagents, and complicates downstream purification. As a result, effective Fe removal is not only essential for improving the efficiency and economics of REE recovery but also represents a critical step toward transforming coal-derived leachates into viable secondary resources for the clean energy future.

Various strategies have been applied for Fe removal from acidic leachates, each with distinct strengths and limitations. Conventional

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neutralization/precipitation—typically achieved by adding lime or alkali—can reduce soluble Fe(III) to insoluble $Fe(OH)_3$. This method is straightforward and relatively low-cost at small scale but requires large reagent inputs and produces significant sludge volumes, particularly at high Fe concentrations, which complicates downstream REE recovery. Although precipitation is effective for bulk Fe removal at high Fe concentrations, it generally requires substantial pH adjustment and may increase reagent consumption, sludge generation, and REE loss through co-precipitation. Ion-exchange and chelating-resin adsorption have also been explored for selective separation of Fe and other impurity metals [8]. These solid-phase approaches offer advantages such as regenerability, reduced secondary waste generation, and compatibility with fixed-bed operation, although their selectivity and capacity are strongly dependent on resin functionality, acidity, and competing ions. Solvent extraction is another widely used method, with di-(2-ethylhexyl) phosphoric acid (D2EHPA) among the most effective extractants for Fe(III) removal. Jin et al. employed D2EHPA in a kerosene-based extraction system to achieve highly efficient Fe(III) removal from hydrochloric acid-route phosphoric acid, reaching extraction efficiencies above 96% through multi-stage cross-current operation [9]. More recently, Zhou et al. demonstrated that saponified D2EHPA could selectively extract Fe from simulated Ni—Cd battery sulfuric acid leachates, achieving single-stage extraction efficiencies above 95% and overall efficiencies greater than 99% after two-stage operation, with Fe recovered as iron oxalate [10]. However, its conventional use in liquid–liquid extraction suffers from significant drawbacks, including high solvent consumption, emulsification and phase-separation issues, and secondary organic waste streams. To overcome these challenges, recent efforts have focused on immobilizing D2EHPA onto solid supports to construct adsorption-type materials [11,12]. In such extraction-type adsorbents, the extractant is immobilized within or on a solid polymeric matrix, allowing metal ions to be selectively complexed by the functional extractant while the material is operated as a solid-phase adsorbent. This configuration combines the coordination chemistry of solvent extraction with the operational advantages of adsorption and fixed-bed processing. This hybrid strategy preserves the strong coordination affinity of D2EHPA toward Fe while simultaneously addressing the shortcomings of liquid–liquid extraction. Compared with conventional solvent extraction, D2EHPA-based adsorbents offer easier operation, minimal solvent loss, improved recyclability, and compatibility with continuous-flow fixed-bed systems.

Although D2EHPA has been used for Fe removal, most studies have emphasized extraction performance rather than the fundamental basis of Fe-over-REE selectivity. The interplay between pH-dependent Fe speciation and ligand deprotonation remains poorly understood, and the role of water molecules in mediating coordination has rarely been considered. Here, using neodymium (Nd) as a representative REE, we integrate spectroscopic characterization (zeta potential, XPS, FTIR) with density functional theory (DFT) to elucidate the coordination pathways of Fe in competitive environments. Particular attention is given to ligand deprotonation, hydrolysis speciation, and water-assisted coordination. Kinetic and isotherm analyses define adsorption behavior under controlled conditions, while fixed-bed column tests assess breakthrough, desorption, and regeneration performance. By coupling molecular-level insights with process-scale validation, this work establishes a unified framework for understanding and optimizing Fe selectivity over REEs from coal-derived acidic leachates.

2. Materials and methods

2.1. Materials

The target solution used in this work was a synthetic coal-derived acidic leachate designed based on the major ionic matrix of a reference leachate generated from Fire Clay coal middlings investigated in our broader project. The Fire Clay middlings contain approximately 200–300 mg/kg total REEs on a whole-mass basis, while the REE-enriched mineral fraction produced after grinding and flotation contains approximately 1000 mg/kg mixed REEs. This mineral-rich fraction was used as the feed for leaching at a slurry concentration of approximately 20 wt%. Owing to the relatively high leachability and low crystallinity of the REE-bearing minerals in the Fire Clay material, no roasting pretreatment was required prior to leaching. The resulting leachate composition provided the basis for preparing the synthetic solution summarized in Table 1, with the initial pH adjusted to approximately 2.0. Nd was employed as a representative trivalent REE for mechanistic investigation of Fe(III)-over-REE selectivity, rather than because it is necessarily one of the most abundant REEs in coal-derived leachates. Its concentration was selected to provide a sufficiently measurable and controlled competitive system for adsorption, selectivity, and molecular-level analyses. The measured redox potential (E_h = +0.51 V versus the standard hydrogen electrode, SHE) confirmed that the simulated leachate remained in an oxidizing state, ensuring that iron existed predominantly as Fe(III). All chemicals used for synthetic solution preparation were purchased from Thermo Fisher Scientific (USA). A commercial D2EHPA-based extraction resin was used as the adsorbent in this study. The material contains di-(2-ethylhexyl) phosphoric acid (D2EHPA) as the functional extractant and was used in the H^+ ionic form. The commercial product name and supplier information are not disclosed due to confidentiality restrictions. Hydrochloric acid (HCl, 35–38 wt%, ACS grade) and sodium hydroxide (NaOH, ≥97 wt% purity) for pH adjustment and adsorbent regeneration were also purchased from Thermo Fisher Scientific. Type I deionized water (resistivity 18.2 MΩ·cm at ambient temperature) was used in all experiments.

2.2. Calculation methods

The aqueous speciation of Fe under different solution conditions and pH values was calculated using Visual MINTEQ 4.0. The software was employed to predict the distribution of Fe species in the synthetic leachate, providing insights into the relative abundance of free Fe^{3+} ions, hydrolyzed complexes (e.g., $FeOH^{2+}$, $Fe(OH)_2^+$), and other relevant species.

With the aim of elucidating the competitive adsorption behaviors of Fe(III) and Nd(III) with D2EHPA, first-principles calculations based on DFT were carried out using the CASTEP program. The exchange-correlation energy was treated within the generalized gradient approximation (GGA) using the Perdew-Burke-Ernzerhof (PBE) functional. A cubic simulation box with a side length of 30 Å was employed, which was sufficiently large to avoid artificial interactions among periodic images, thus eliminating the need for an additional vacuum layer. The plane-wave basis set was truncated at an energy cutoff of 280 eV, and the electronic self-consistent field convergence criterion was set to 1×10^{-5} eV.

The adsorption energy (E_{ads}) was defined as:

$$E_{ads} = E_{complex} - \left(E_{ligand} + E_{ion}\right) \quad (1)$$

where $E_{complex}$ represents the total energy of the D2EHPA-ion system, E_{ligand} is the energy of the isolated D2EHPA molecule, and E_{ion} is the energy of the individual metal ion.

2.3. Adsorption and column experiments

Batch adsorption experiments were conducted to evaluate the uptake capacity of the adsorbent toward Fe (or REEs). All experiments were performed in an orbital shaker (120 rpm) under ambient conditions with a fixed adsorbent-to-solution ratio of 30 g/150 mL unless otherwise specified. Adsorption capacity (*Eq.*(2)) and removal efficiency (*Eq.*(3)) were determined by comparing the ion concentrations in solution before and after reaction. For pH-dependent adsorption tests, the solution pH was manually adjusted to 2.0, 2.5, or 3.0, while keeping other

Table 1
Composition and properties of the synthetic coal-derived acidic leachate.

Unit	Nd	Fe(III)	Al	Ca	Mg	Li	Ni	Co	Mn	Sr	Zn	Eh
mg/L	70	3000	700	400	100	30.91	13.66	7.50	5.54	8.61	22.35	+0.51 V

parameters constant. For pH control, standardized solutions of HCl (1.0 and 0.10 M) and NaOH (1.0 and 0.10 M) were used and added dropwise (total addition ≤1% *v*/v).

$$q = \frac{v \times (C_0 - C_f)}{m} \tag{2}$$

$$E = \frac{100\% \times (C_0 - C_f)}{C_0} \tag{3}$$

where C_0 (mg/L) and C_f (mg/L) refer to the Fe (or REE) ion concentration in the synthetic acidic leachate solution before and after adsorption, respectively; v (L) is the volume of the feed solution, and m (g) is the added mass of the adsorbent material.

Kinetic experiments were carried out by sampling aliquots at predetermined time intervals to monitor the adsorption process. The kinetic data were fitted using both pseudo-first-order (*Eq.*(*S*1)) and pseudo-second-order (*Eq.*(*S*2)) models to identify the dominant adsorption mechanism. Adsorption isotherms were obtained at room temperature with different adsorbent loadings (0.5, 1, 2, 5, 10, 30, and 50 g per 150 mL). The results were further fitted with Freundlich (*Eq.*(*S*3)) and Langmuir (*Eq.*(*S*4)) models to derive the characteristic adsorption parameters and evaluate the monolayer versus heterogeneous adsorption behavior. To evaluate the selectivity of the material toward Fe, the following equations were applied to calculate the distribution coefficient k_{Me} and the separation factor a_{Me}^{Fe}:

$$k_{Me} = \frac{(C_0 - C_e) \bullet v}{C_e \bullet m} \tag{4}$$

$$a_{Me}^{Fe} = \frac{K_{Fe}}{K_{Me}} \tag{5}$$

where C_0 (mg/L) and C_e (mg/L) refer to the initial ion concentration and the equilibrium concentration, respectively, m (g) is the mass of adsorbent added, and v (L) is the solution volume. The subscript M_e of K_{Me} and a_{Me}^{Fe} refers to the metal ions in the solution.

To evaluate the performance of continuous operation and simulate cyclic application, adsorption–desorption column experiments were carried out. Approximately 210 g of adsorbent were packed into a glass chromatography column with dimensions of 10 mm in diameter and 30 cm in height, followed by adsorption–desorption tests. During the adsorption process, the effluent concentration was monitored at fixed volumes to construct breakthrough curves and evaluate the dynamic adsorption behavior. Three empirical models—the Modified Dose–Response model, the Clark model, and the Yoon–Nelson model—were further applied to simulate and analyze the adsorption process. Detailed descriptions of these models are provided in the Supporting Information.

For the desorption experiment in the column system, the Fe desorption efficiency D_e (%) in the desorption stage can be employed to evaluate the Fe release performance. The calculation was followed by *Eq.*(6).

$$D_e = 100\% \times \frac{v \bullet C_d}{q_{total}} \tag{6}$$

where v (L) refers to the volume of the total desorption effluent at time t, C_d (mg/L) means the Fe concentration in the total desorption effluent at time t, and q_{total} (mg) is the total amount of Fe adsorbed in the adsorption stage, which can be calculated using *Eq.*(*S*11). Finally, adsorption–desorption cycling experiments were carried out for eight successive cycles to further assess the long-term stability and regenerability of the adsorbent under fixed-bed conditions. All the above experiments were conducted three times to ensure good repeatability.

2.4. Assays

To comprehensively evaluate the physicochemical properties of the adsorbent, a series of analytical techniques were employed. The surface charge characteristics were determined by zeta potential analysis using a Zetasizer Nano ZS (Malvern Instruments, UK), providing insight into interfacial stability under varying pH conditions. Ion concentrations in solution were quantified by inductively coupled plasma mass spectrometry (ICP-MS, Thermo Fisher Scientific, USA), enabling precise determination of Fe, Nd, and other competing ions. X-ray photoelectron spectroscopy (XPS) was carried out on a PHI Quantera Hybrid instrument equipped with a monochromatic Al Kα source (1486.6 eV); all spectra were calibrated to the C 1 s peak at 284.8 eV to assess surface composition and metal–ligand coordination environments. Fourier transform infrared spectroscopy (FTIR) measurements were performed using KBr pellets to probe the functional groups involved in adsorption and desorption.

3. Results and discussion

3.1. Molecular-level insights into Fe selectivity over Nd

To assess the binding selectivity of D2EHPA, the removal behaviors of Fe and Nd were compared (Fig. 1A). In the early stage (2–5 min), both Fe and Nd were extracted to a considerable extent. As the reaction proceeded, however, Nd removal rapidly declined and stabilized at a low level, whereas Fe removal continued to increase and eventually approached complete extraction. This pronounced divergence reveals the stronger affinity and sustained complexation capacity of D2EHPA toward Fe^{3+} relative to Nd^{3+}, demonstrating its high selectivity for Fe removal. To further illustrate the pH-dependent competitive behavior, the Fe and Nd removal efficiencies after 120 min were compared at different pH values (Fig. S1). Fe remained preferentially removed at pH 2.0–2.5, whereas increased Nd co-adsorption at pH 3.0 resulted in reduced Fe-over-Nd selectivity.

To elucidate the molecular origin of this selectivity, theoretical calculations were conducted (Fig. 1C–E). In single-component systems, Fe–D2EHPA exhibits markedly stronger interaction energies than Nd–D2EHPA, indicating more stable Fe coordination with the extractant. When Fe, Nd, and D2EHPA coexist, the binding energies of both complexes decrease due to competitive ion interactions; nevertheless, Fe–D2EHPA retains higher stability, contributing to Fe's preferential adsorption and the progressive displacement of Nd during the process. Charge transfer analysis (Fig. 1D) further supports this conclusion: Fe shows greater electron accumulation (ΔQ ≈ +0.9 e) than Nd (ΔQ ≈ +0.7–0.8 e), confirming its stronger electron-accepting ability. ProjectedPartial density of states (PDOS) analysis (Fig. 1E) reveals pronounced orbital overlap between Fe 3d and O 2p orbitals near the Fermi level (−5 to 0 eV), signifying strong orbital coupling and hybridization that reinforce Fe—O coordination. In contrast, the overlap between Nd 4d/4f orbitals and ligand orbitals is much weaker, indicating limited electronic interaction with D2EHPA.

Based on these findings, we propose a molecular-level adsorption pathway (Fig. 1B). Under acidic conditions, D2EHPA primarily exists in its protonated form –P(O)(OH)–OR, in which the hydroxyl group cannot effectively coordinate Fe^{3+}. Upon interaction with Fe^{3+}, D2EHPA undergoes partial deprotonation to form negatively charged ligands ($–O^-$),

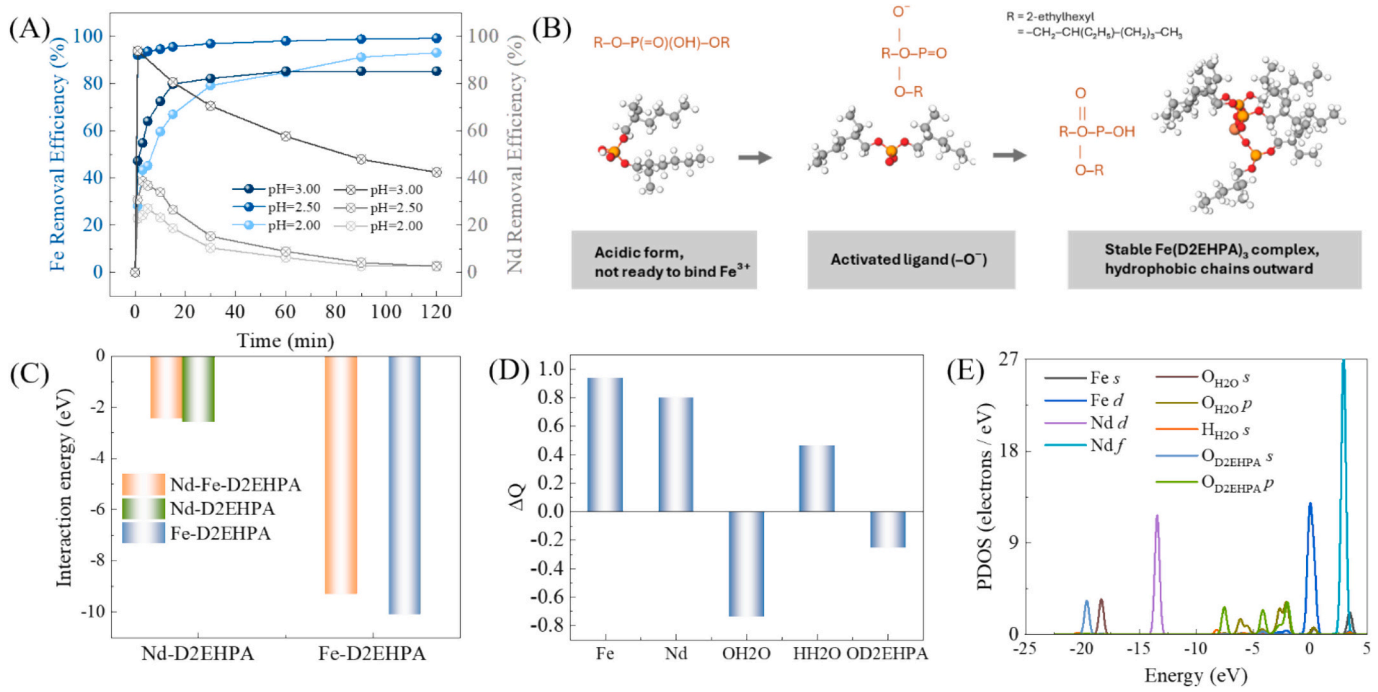


Fig. 1. Mechanistic and computational insights into selective Fe removal using D2EHPA. (A) Time-dependent Fe and Nd extraction efficiencies at different pH values. (B) Schematic of the Fe–D2EHPA complexation mechanism (Atom colors: Gray = C; White = H; Red = O; Orange = P). (C) Interaction energies from DFT calculations. (D) Charge transfer analysis (ΔQ) of the Fe–Nd–D2EHPA system. (E) Partial density of states (PDOS) spectra for Fe, Nd, water, and D2EHPA. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

facilitating stronger electrostatic and coordinative interactions. Multiple deprotonated D2EHPA molecules then bind Fe^{3+} to form stable coordination complexes, while their long alkyl chains extend outward to create a hydrophobic shell. This structural arrangement not only stabilizes the Fe–D2EHPA complex but also hinders competing ions from binding. In addition to this ligand-driven pathway, the DFT results (Fig. 1C and D) also highlight the cooperative role of water molecules. Coordinated water oxygen atoms donate electrons to Fe^{3+}, while their hydrogen atoms form hydrogen bonds with D2EHPA oxygen atoms. These dual interactions facilitate electron transfer, shorten the effective Fe–O(D2EHPA) distance, and further stabilize Fe binding. Importantly, water molecules act only as mediators that assist coordination and electron redistribution during the process, rather than remaining in the final Fe–ligand complex. The PDOS analysis (Fig. 1E) confirms strong orbital overlap between Fe d orbitals and D2EHPA O p orbitals with additional contributions from water oxygen states, collectively reinforcing Fe–ligand coordination. This combined mechanism—ligand deprotonation, water-assisted coordination, and orbital coupling—explains the enhanced stability of Fe–D2EHPA complexes over Nd–D2EHPA, accounting for the progressive and nearly complete removal of Fe during the competitive adsorption process.

3.2. *pH-gated Fe(III) speciation and D2EHPA deprotonation behavior*

The solution pH simultaneously regulates the deprotonation state of D2EHPA and the distribution of Fe(III) hydrolysis species, thereby jointly determining the coordination mode and adsorption strength of Fe. As shown in Fig. 2A, the Fe adsorption capacity increases with pH and reaches a maximum of 12.1 mg/g at pH ≈ 2.5, followed by a slight decline at pH 3.0. This distinct pH dependence arises from the coupled effects of Fe speciation and ligand deprotonation. Speciation analysis (Fig. 2B) shows that Fe^{3+} predominates (>70%) at pH ≈ 2.0, whereas $FeOH^{2+}$ concentration increases sharply with pH, becoming comparable to Fe^{3+} near pH ≈ 2.9. The coexistence of abundant Fe^{3+} and emerging $FeOH^{2+}$ species at pH ≈ 2.5 yields the highest adsorption, suggesting synergistic contributions from both species. Thus, both Fe speciation and D2EHPA deprotonation are jointly responsible for the observed pH dependence.

The zeta potential and IEP data (Fig. 2C and D) further corroborate the effect of pH on interfacial charge and binding behavior. Compared with the raw material, all Fe-adsorbed samples show IEP shifts and markedly decreased zeta potentials, indicating specific adsorption of Fe onto surface active sites rather than mere electrostatic attraction. The IEP shifts positively after Fe adsorption and reaches a maximum (~2.0) at pH ≈ 2.5, consistent with more effective charge neutralization via Fe–ligand interactions. In contrast, the IEP remains low at pH 2.0 due to insufficient D2EHPA deprotonation and drops sharply at pH 3.0 likely because of Fe hydrolysis and partial colloidal $Fe(OH)_3$ formation (Fig. S2), which reduce soluble Fe availability and surface complexation. These electrokinetic results imply that beyond electrostatic attraction, specific coordination interactions are involved in Fe binding.

Spectroscopic analyses were conducted to provide direct molecular-level evidence of Fe—O(P) coordination. As shown in Fig. S3(A), the XPS survey spectra display characteristic C 1 s, O 1 s, and P 2p peaks of the adsorbent, while distinct Fe 2p signals appear only in Fe-adsorbed samples (pH 2.0–3.0) and are absent in the raw material, confirming successful Fe uptake.

High-resolution P 2p spectra (Fig. 2E) of the raw material reveal four types of P—O environments, including P═O, P—OH, and P—O—C bonds, together with a small fraction of hydrogen-bonded P—O species. Upon Fe adsorption, a new peak emerges at ~134.5 eV corresponding to P—O—Fe coordination [13], while the P—OH contribution decreases and the remaining peaks shift toward higher binding energies, indicating ligand deprotonation and electron withdrawal upon Fe binding [14]. The P—O—Fe component is most pronounced at pH ≈ 2.5, coinciding with the strongest inner-sphere coordination and maximum Fe uptake.

The Fe 2p ₃/₂ spectra (Fig. 2F) further substantiate this conclusion. Distinct Fe^{3+} peaks at ~710–711 eV with satellites near 713–715 eV appear exclusively after adsorption, with maximum intensity at pH ≈

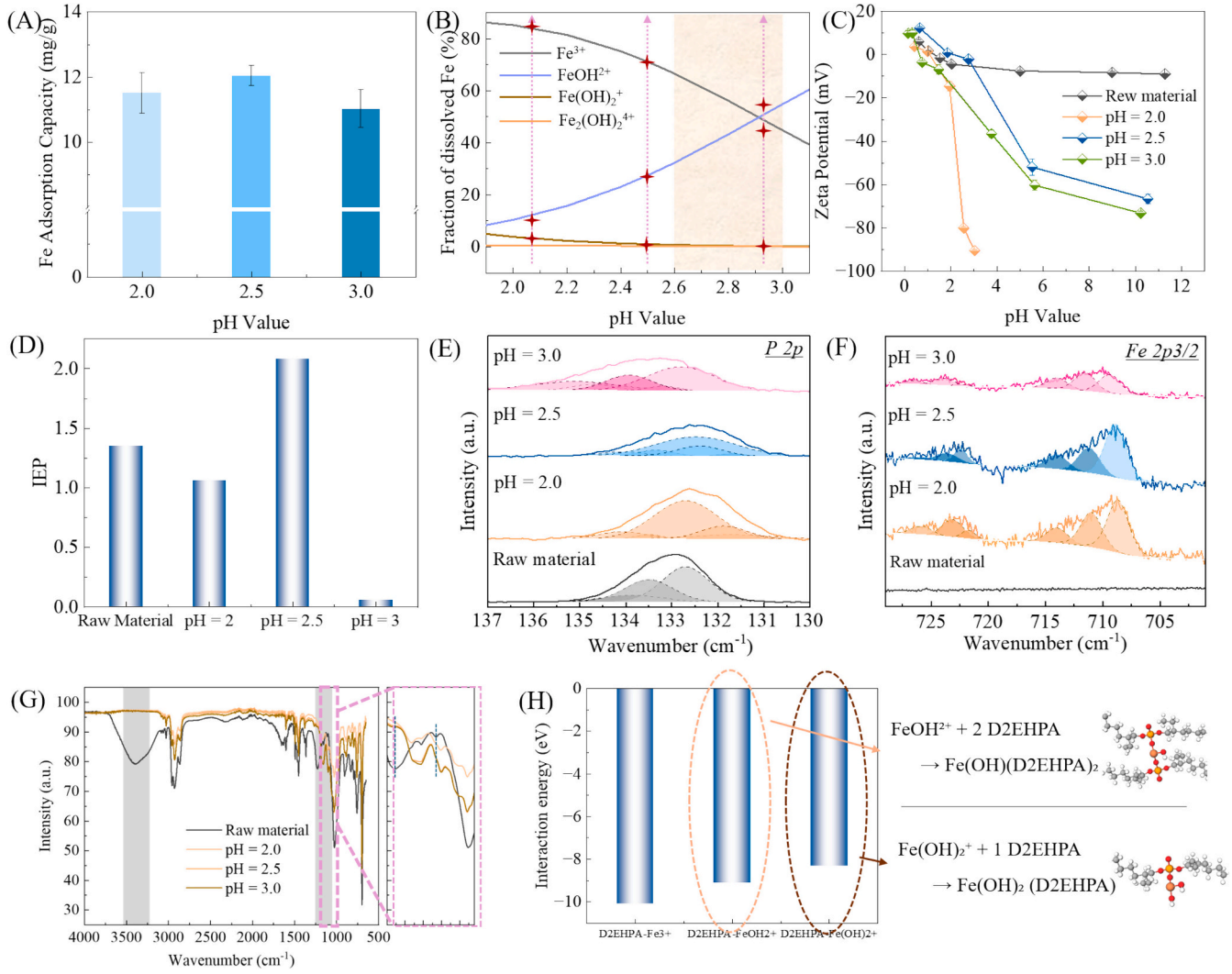


Fig. 2. (A) Fe adsorption capacity at different pH values. (B) Distribution of Fe species as a function of pH (qualitative illustration, Cl^- complexes and solid phases not considered). (C) Zeta potential profiles of the raw material and after Fe adsorption at different pH. (D) IEP shifts upon Fe adsorption. (E, F) High-resolution XPS spectra of P 2p and Fe 2p3. (G) FTIR spectra of raw material and samples after adsorption at different pH. (H) Proposed binding mechanisms of $FeOH^{2+}$ and $Fe(OH)_2^+$ with D2EHPA molecules, along with DFT-calculated interaction energies.

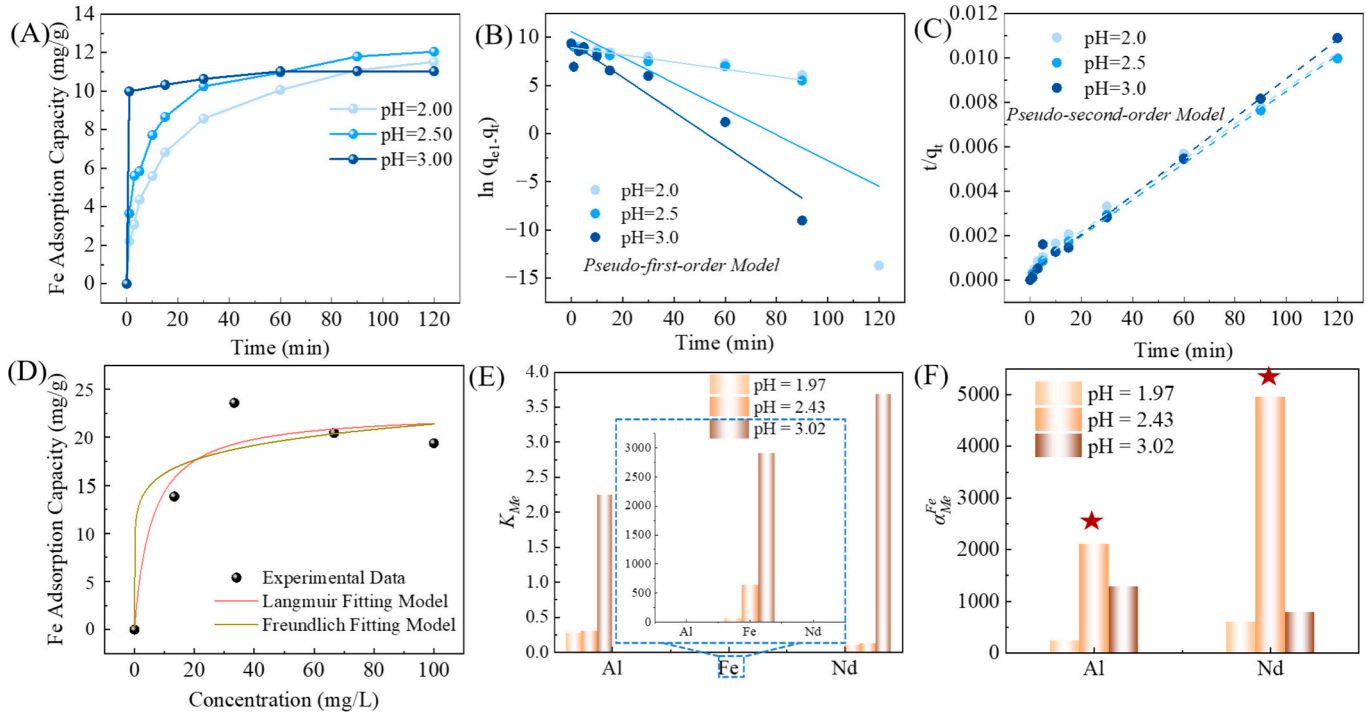
2.5, signifying greater surface coverage and stronger coordination. The signal weakens at pH 2.0 due to limited D2EHPA deprotonation and diminishes again at pH 3.0 owing to Fe^{3+} hydrolysis and partial $Fe(OH)_3$ formation. Similarly, high-resolution O 1 s spectra (Fig. S3(B)) exhibit shifts toward higher binding energies and increased Fe—O (~530–531 eV) intensity, confirming the participation of oxygen species (P—O and O—Fe) in complexation.

Complementary FTIR spectra (Fig. 2G) further reinforce these observations. The broad band at ~3400 cm^{-1} assigned to O—H stretching of hydroxyls and hydrogen-bonded water diminishes significantly after Fe adsorption, indicating deprotonation or replacement of —OH and P—OH groups. The characteristic bands at ~1230 cm^{-1} (P═O asymmetric stretching) [15] and ~1010 cm^{-1} (P–OH/P–O vibrations) [16] also weaken and shift, consistent with the formation of Fe—O—P linkages. Together, these XPS and FTIR results unambiguously confirm the formation of inner-sphere Fe—O(P) coordination bonds.

DFT calculations (Fig. 2H) further support this mechanism, showing that Fe^{3+} and $FeOH^{2+}$ exhibit the most favorable binding interactions with D2EHPA, with calculated binding energies of −10.09 eV and −9.09 eV, respectively. The optimized structure of the $FeOH^{2+}$–$(D2EHPA)_2$ complex reveals a neutral inner-sphere coordination configuration, where the Fe center is surrounded by six oxygen atoms from one hydroxyl group and two deprotonated D2EHPA ligands, forming an octahedral geometry with Fe—O bond distances in the typical range of 1.9–2.1 Å and O–Fe–O angles of approximately 90° and 180° [17]. Considering that $FeOH^{2+}$ is the predominant hydrolyzed species at pH ≈ 2.5, this octahedral $Fe(OH)(D2EHPA)_2$ complex likely represents the dominant coordination structure during adsorption. At higher pH, the conversion of Fe^{3+} to $Fe(OH)_3$ reduces the availability of reactive Fe species and weakens inner-sphere coordination, resulting in decreased Fe uptake. Collectively, the experimental and theoretical results establish pH-regulated inner-sphere Fe–O(P) coordination as the governing mechanism for Fe removal, with pH ≈ 2.5 representing the optimal condition for selective complexation.

3.3. Fe adsorption behavior: Kinetics, isotherms, and selectivity

The Fe adsorption kinetics of the material were examined in greater detail. As shown in Fig. 3A, Fe uptake was rapid within the first 20 min and gradually approached equilibrium. Under pH 2.0–2.5 conditions, adsorption continued to increase slowly, reaching equilibrium after ~90 min. In contrast, at pH 3.0, adsorption reached apparent equilibrium within ~30 min, much faster than at lower pH. However, the final capacity was smaller, which may reflect reduced Fe^{3+} availability and



(G)

pH	Qexp. (mg/g)	Pseudo-first-order Model			Pseudo-second-order Model		
		k_1 (1/min)	q_{e1} (mg/g)	R^2	k_2 (g/(mg·min))	q_{e2} (mg/g)	R^2
2.0	12.01465	0.13398	40.04573	0.65296	1.11E-05	12.4	0.99154
2.5	12.04310	0.03777	7.68518	0.65296	3.51E-04	28.5	0.9968
3.0	11.02967	0.17928	12.51999	0.65296	2.56E-05	11.4	0.99118

Fig. 3. (A) Time-dependent adsorption capacity of Fe at different pH values. (B, C) Kinetic model fitting. (D) Adsorption isotherms fitted by Langmuir and Freundlich models. (E, F) Distribution and selectivity factors of Fe relative to Al and Nd, respectively, showing distinct pH dependence.

partial hydrolysis rather than an intrinsically higher adsorption rate. The rapid initial increase likely corresponds to the quick surface complexation of the remaining soluble Fe species, followed by early saturation [18]. To elucidate the adsorption mechanism, kinetic modeling was performed (Fig. 3B and C). The pseudo-second-order (PSO) model fitted the data excellently ($R^2 > 0.99$), clearly outperforming the pseudo-first-order model, indicating that Fe adsorption is dominated by chemisorption via inner-sphere coordination—consistent with the spectroscopic findings [19]. As shown in Fig. 3G, the PSO rate constant k_2 at pH 2.5 (3.51×10^{-4} g/(mg·min)) exceeds those at pH 2.0 and 3.0, identifying pH 2.5 as the kinetic optimum. These results confirm that Fe uptake at pH 2.5 proceeds with faster intrinsic kinetics and stronger coordination, aligning with the thermodynamic and spectroscopic trends.

To further probe the interfacial adsorption behavior, Fe adsorption isotherm was investigated, as well as the fitting models (Fig. 3D). The results showed that the Langmuir model provided a better fit than the Freundlich model, indicating that Fe adsorption predominantly follows a monolayer process on relatively homogeneous binding sites. The Langmuir constant $b = 0.17068$ L/mg and the separation factor $R_L = 0.002122$ (< 1) indicate highly favorable uptake over the investigated concentration range [20]. In contrast, the poorer fit of the Freundlich model suggests limited surface heterogeneity and the absence of pronounced multilayer adsorption. Taken together with the PSO kinetics and spectroscopic evidence, these results support that Fe uptake proceeds predominantly via monolayer chemisorption on uniform surface sites. Furthermore, as shown in Fig. 3E, the distribution coefficient of Fe was consistently much higher than that of Al and Nd across all pH conditions, demonstrating the significantly stronger adsorption affinity of the material for Fe. Notably, K_{Fe} reached its maximum at pH = 3.0, indicating that the adsorption driving force for Fe was strongest under this condition. However, the selectivity coefficients (Fig. 3F) revealed that the separation selectivity of Fe relative to Al and Nd was highest at pH = 2.5, where both α_{Al}^{Fe} and α_{Nd}^{Fe} reached values on the order of 10 [3]. This suggests that in competitive adsorption systems, pH = 2.5 is more favorable for the preferential removal of Fe. This difference can be attributed to Fe speciation and coordination mechanisms. At pH ≈ 2.5, the coexistence of abundant Fe^{3+} and emerging $FeOH^{2+}$ species facilitates adsorption, where $FeOH^{2+}$ forms electroneutral 2:1 inner-sphere complexes [Fe(OH)(D2EHPA)₂] that dominate selective Fe removal. These charge-balanced complexes not only ensure high Fe uptake but also suppress the competitive adsorption of Al and Nd, resulting in superior selectivity. In contrast, at pH = 3.0, although K_{Fe} is higher, the sharp decline of Fe^{3+} and advanced hydrolysis weaken effective inner-sphere coordination and reduce competitive selectivity. Taken together, pH 3.0 maximizes the apparent distribution coefficient, whereas pH 2.5 is optimal for selective Fe removal. Collectively, these results confirm that Fe adsorption on D2EHPA proceeds via pH-tuned chemisorption on uniform active sites, enabling high capacity and strong selectivity under competitive conditions.

3.4. Dynamic column performance and regeneration stability

To evaluate adsorption performance under dynamic conditions, fixed-bed column experiments were conducted to simulate continuous operation and validate process scalability. As illustrated in Fig. 4A, the feed solution was introduced from the bottom and passed upward through the adsorbent-packed column, where Fe ions were continuously captured (I. Adsorption). Once saturation was approached, a rinsing step (II. Rinsing) stabilized the bed and removed residual liquid, followed by acid elution (III. Desorption) to release the adsorbed ions. After each adsorption–desorption cycle, the adsorbent remained in place, allowing multi-cycle operation and assessment of stability and regenerability.

The effects of bed height and flow rate on Fe breakthrough were systematically examined. As shown in Fig. 4B and C, increasing the bed height-to-diameter (H/D) ratio significantly delayed breakthrough, shifting the curves toward higher bed volumes. Taller beds provided longer mass-transfer paths and more active sites, enhancing capacity and delaying saturation, whereas shorter beds led to early breakthrough. Conversely, increasing the influent flow rate advanced the breakthrough curves: at 5.14 mL/min, Fe penetrated the bed rapidly due to shortened residence time and limited diffusion, while at 1.80 mL/min, extended contact time promoted more efficient adsorption. Effluent monitoring (Fig. S4) further confirmed that higher H/D and lower flow rates effectively suppressed Fe concentrations, ensuring stable and sustained removal.

The breakthrough behavior of Nd was also monitored for comparison. Under varying H/D ratios (Fig. 4D), Nd concentration in the effluent rapidly increased within the first 30–40 min and then stabilized at a high level, largely independent of bed height, indicating weak retention. Under different flow rates (Fig. 4E), higher flows caused faster breakthrough and higher steady-state Nd concentrations, while lower flows slightly reduced the effluent concentration (~60 mg/L). Overall, Nd was poorly retained, with effluent concentrations approaching the influent level. These results clearly demonstrate the material's strong preference for Fe over Nd in continuous-flow systems, confirming its high selectivity and robust dynamic adsorption performance.

The fixed-bed breakthrough profiles were analyzed using three adsorption models (Clark, MDR, and Yoon–Nelson) as shown in Figs. 4F–H. Among them, the Clark model provided the best overall fit ($R^2 > 0.99$) across all bed heights (Table S2), accurately capturing the self-similar breakthrough fronts and reflecting the combined effects of surface heterogeneity and external mass transfer. The resulting parameters indicate improved mass-transfer utilization and higher effective bed capacity at larger H/D ratios [21], supporting the applicability of the Clark framework for scale-up [22].

Additionally, column tests using an actual acid mine drainage further

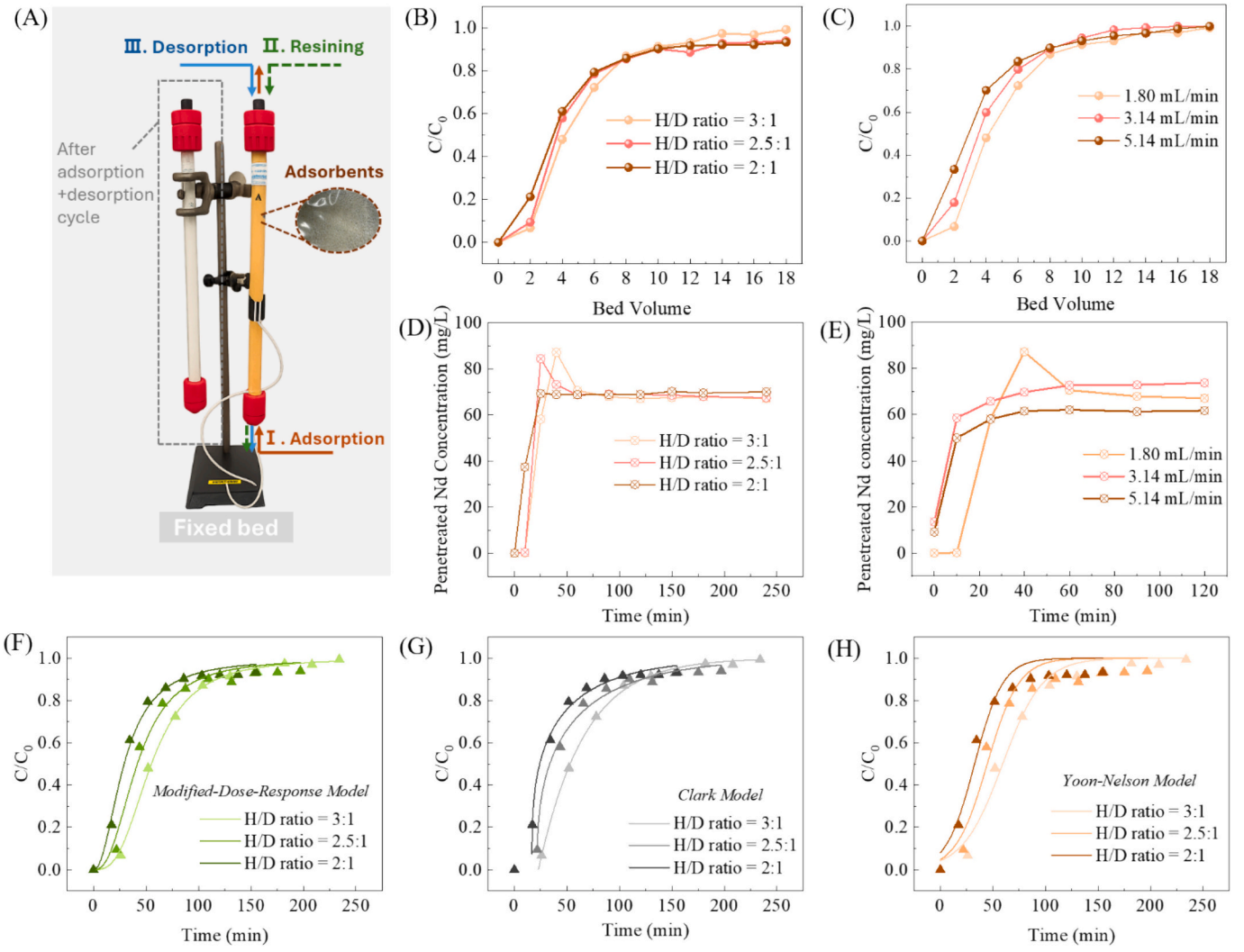


Fig. 4. (A) Schematic of the fixed-bed system. (B, C) Breakthrough curves as a function of H/D ratio (fixed flow rate: 1.8 mL/min) and flow rate (fixed H/D ratio: 3:1), respectively. (D, E) Penetrated Nd concentrations under varying H/D ratios and flow rates, respectively. (F–H) Breakthrough curve fitting using the Clark model, Yoon–Nelson model, and modified dose–response model.

confirmed preferential Fe retention over Y under a complex aqueous matrix (Fig. S5), supporting the applicability of the proposed Fe-removal process beyond the synthetic Fe—Nd system, although competitive effects among individual REEs may still vary with solution composition.

To evaluate regeneration under practical conditions, desorption column tests were performed following Fe loading to examine the effects of acid concentration, flow rate, and co-desorption of Nd. As shown in Fig. 5A and C, Fe desorption is strongly governed by HCl concentration. Within 5–20 wt% HCl, the effluent Fe concentration rises sharply at the initial stage and gradually stabilizes, indicating effective release of adsorbed Fe. The desorption efficiency curves further show that increasing acid strength markedly enhances Fe elution, with the highest efficiency at 20% HCl. This behavior reflects protonation of P—O groups and disruption of Fe—O(P) inner-sphere coordination at high [H^+] (Fig. S6(A)). In contrast, 5% HCl yields only partial desorption, leaving substantial Fe on the bed. These observations confirm that acid concentration dictates both the rate and extent of desorption; however, excessively strong acids may compromise material stability and must be balanced against corrosion concerns in practical operation.

The effect of flow rate is summarized in Fig. 5B and D. Under all conditions, Fe concentration in the effluent initially increases and then levels off, but the overall desorption efficiency differs markedly. At 1.80 mL/min, efficiency remains below 10%, indicating a film-diffusion–limited regime where slow convective renewal causes local acid depletion and thick boundary layers. Increasing the flow to 3.14 and 5.14 mL/min significantly enhances efficiency (up to >80%), attributable to rapid liquid-film renewal and improved external mass transfer (Fig. S6B) [23]. Thus, moderate flow rates are favorable for efficient Fe elution, whereas very low flow severely restricts performance.

Co-desorption of Nd was minimal (Fig. 5E and F). Nd release increased slightly at moderate acidity (10 wt% HCl) and higher flow rates but remained far below Fe, indicating that Fe-over-REE selectivity was largely preserved during regeneration (Figs. S6(C) and S6(D)). Together, these results demonstrate that regeneration is efficient and selective under optimized conditions, confirming the robustness and reusability of the adsorbent in continuous operation.

Column adsorption–desorption cycling is essential for assessing regenerability and long-term performance under industrially relevant conditions. As shown in Fig. 5G, adsorption and desorption masses were tracked over eight consecutive cycles. The first two cycles exhibited attenuated capacities, consistent with an initial bed conditioning/activation period during which wetting, pore equilibration, and exposure of buried sites occur [24]. Thereafter, the system reached a steady state: from cycle 3 onward, the adsorption stabilized at ~375 mg per cycle, while desorption remained within 250–300 mg per cycle, with no

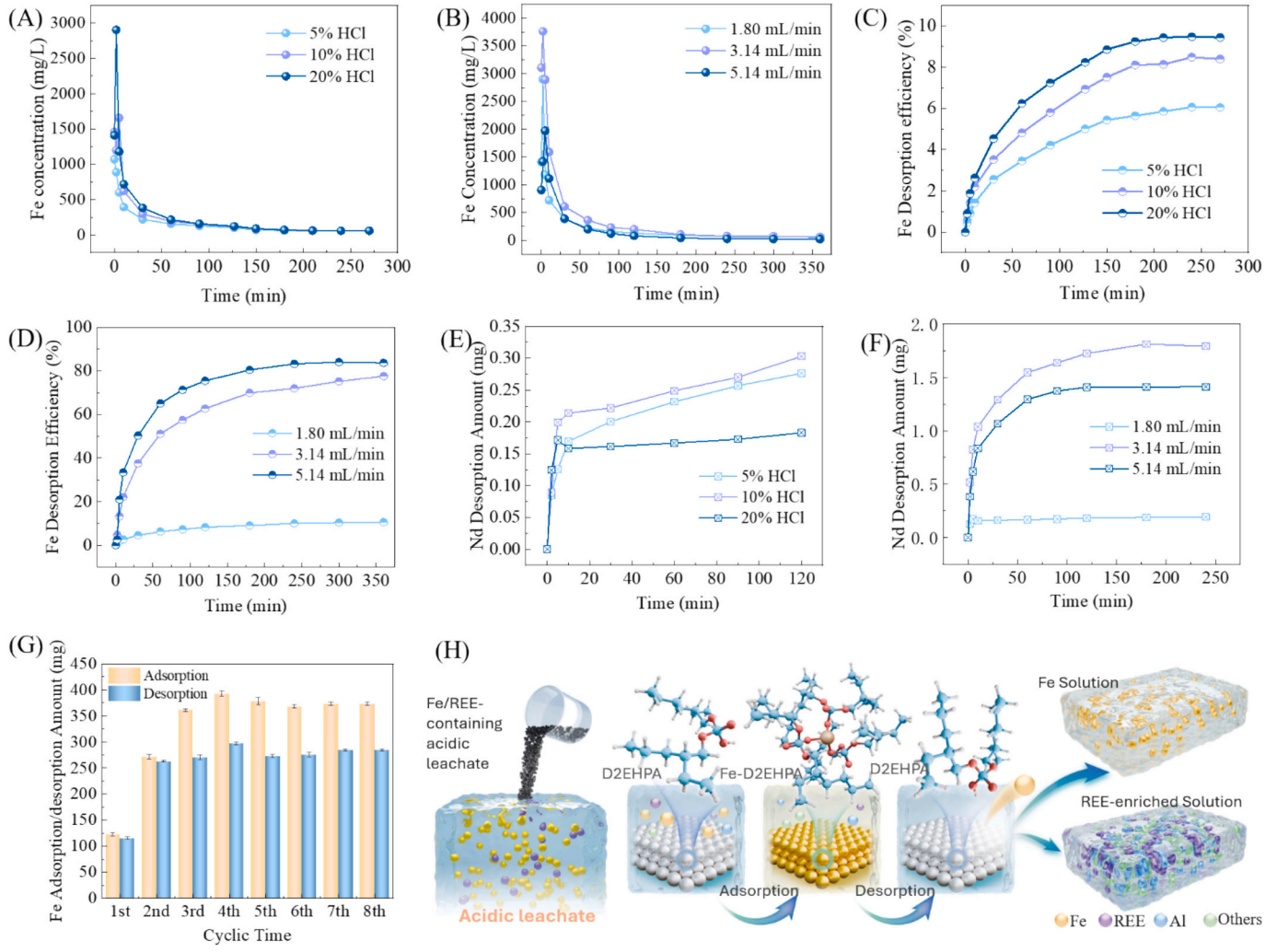


Fig. 5. (A, B) Effluent Fe concentrations under different HCl concentrations and flow rates, respectively. (C, D) Fe desorption efficiency as a function of HCl concentration and flow rate. (E, F) Nd desorption amounts under different HCl concentrations and flow rates. (G) Adsorption–desorption cycling stability of Fe over eight consecutive runs (One cycle comprises one adsorption, one rinsing, and one desorption step). (H) Schematic illustration of Fe adsorption and desorption by a D2EHPA-based resin from representative Fe/REE-containing acidic leachates (e.g., coal-related sources).

discernible downward trend. The persistent gap between adsorption and desorption reflects a small residual loading retained after each elution—common for inner-sphere sites—and could be further reduced by stronger eluent or extended rinsing if desired. Overall, these data demonstrate excellent structural integrity and regenerability of the sorbent under fixed-bed operation and show that, after a brief activation stage, the column maintains stable adsorption–desorption performance over repeated cycles.

Fig. 5H schematizes the D2EHPA-enabled route for preferential Fe removal and REE enrichment. In acidic coal-derived leachates containing abundant Fe and trace REE, selective Fe pre-removal is a prerequisite for efficient REE recovery. During adsorption, deprotonated D2EHPA forms inner-sphere Fe–O(P) complexes (2:1 with Fe^{3+}/$FeOH^{2+}$), capturing Fe, while REE—exhibiting much weaker affinity under these conditions—passes through the bed, yielding an Fe-depleted, REE-enriched raffinate. During desorption, HCl protonates P—O and cleaves Fe—O(P) bonds, producing an Fe-rich eluate from the column. The two-stream outcome (Fe-rich eluate vs Fe-depleted/REE-enriched raffinate) demonstrates effective Fe–REE separation and strong Fe selectivity of D2EHPA-based sorbents under complex acidic matrices, providing a practical front-end step for REE recovery from coal-derived leachates.

4. Conclusions

This study establishes a mechanistic framework for selective Fe(III) removal from acidic REE-containing leachates using a D2EHPA-based resin. The results demonstrate that Fe-over-REE selectivity is governed by the coupled effects of Fe(III) aqueous speciation and D2EHPA deprotonation. An optimum adsorption condition was observed at approximately pH 2.5, where the coexistence of Fe^{3+} and hydrolyzed $FeOH^{2+}$ species favors strong coordination with deprotonated D2EHPA. Spectroscopic analyses further confirmed the formation of inner-sphere Fe—O(P) bonds, while DFT calculations showed that Fe–D2EHPA complexes exhibit stronger interactions and greater charge transfer than the corresponding Nd complexes. These results collectively identify specific coordination, rather than simple electrostatic attraction, as the molecular origin of preferential Fe uptake.

The adsorption behavior further supports this mechanism. Fe uptake was well described by the pseudo-second-order kinetic and Langmuir isotherm models, indicating predominantly chemisorption-controlled monolayer adsorption on relatively uniform active sites. Under competitive conditions, the resin exhibited substantially higher affinity and selectivity toward Fe than toward Nd and Al, with the strongest Fe-over-REE selectivity observed near pH 2.5. Continuous-flow column experiments further demonstrated effective Fe retention, limited Nd uptake, efficient acid-assisted regeneration, and stable adsorption–desorption performance over repeated cycles.

Overall, this work goes beyond demonstrating Fe removal efficiency by providing a molecular-level explanation for the selectivity of D2EHPA-based adsorbents toward Fe(III) over REEs. The combination of mechanistic understanding and fixed-bed validation provides a basis for rational optimization of Fe pre-removal prior to downstream REE concentration and purification. The proposed approach therefore offers a practical route for reducing Fe interference while minimizing REE losses in complex acidic leachates. Further evaluation using multi-REE systems and additional real leachates will be valuable for establishing the broader applicability of the process.

Beyond coal-derived leachates, the Fe-over-REE selectivity identified in this study may also be relevant to other Fe-rich REE recovery systems, particularly leachates generated from end-of-life NdFeB permanent magnets. Such leachates contain substantial amounts of Fe together with Nd and other REEs, making selective Fe pre-removal an important step before downstream REE separation and purification. The preferential Fe–D2EHPA coordination demonstrated here therefore provides a mechanistic basis for extending this approach to NdFeB recycling and related REE-containing process streams.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.seppur.2026.140463.

Data availability

Data will be made available on request.

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